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ORIGINAL RESEARCH

Design for Extended Reality (DfXR): A Framework for Educational Services

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Abstract

The rapid adoption of extended reality (XR) is driving the development of immersive educational services and increasing the need for systematic design approaches. From a Design for Extended Reality (DfXR) perspective, this study investigates authentic XR use-cases to develop an empirically grounded framework that synthesises design principles supporting educational service innovation, learner engagement, user adoption, and organisational implementation. The study is based on qualitative findings from open-ended provider and user-reflections, structured queries where use-case descriptions have been gathered. It explores a series of authentic use-cases, delving into both design and user experience aspects. The data was collected through firsthand experimental testing, use-case workshop reflections, and insights from the lead designer. In addition, two technology providers added use-case insights together with two classes of engineering design students. The findings uncover use-case design insights and their value concerns. Past research has shown that XR service solutions deliver added value to various stakeholders, employees, and customers on multiple fronts. The study identifies critical aspects for establishing user adoption proficiency and highlights that delivery value deviates between stakeholders, benefits, and practical use. This study contributes to the emerging DfXR discourse by synthesising findings from authentic XR use-cases into an empirically grounded framework for educational service design. The proposed framework demonstrates how pedagogical, technological, and organisational design considerations can be systematically integrated to support immersive learning, user adoption, and organisational implementation. By shifting the focus from technological capability to intentional design, the study provides practical guidance for developing scalable and effective XR educational services across higher education and professional learning contexts.

Keywords Use-case · Immersive learning · XR · Educational services

Introduction

Extended Reality (XR), encompassing virtual reality (VR), augmented reality (AR) and mixed reality (MR), has emerged as a transformative technology capable of supporting immersive, interactive and experiential learning environments across higher education and professional training. Through increased learner presence, contextualised interaction and authentic simulation, XR enables educational experiences that extend beyond traditional classroom delivery while promoting learner engagement and knowledge retention [1–3]. Today, XR have presented itself as an impactful tool for establishing immersive user experiences, enabling circular business services and various immersive environments to drive sustainability and pursue SDG targets [4, 5]. Past research indicates that the time needed to develop effective new XR scenarios is a crucial factor for the broader adoption of these technologies in the industry [6]. Technology providers present a new path for digitalised service offerings, with the growing number of applications that enhance consumer experience and engagement [7]. The service interface is increasingly evolving towards a more technology-driven setup, incorporating emerging technologies based on accelerated engagement, and increased maturity level to technological means [8]. Recent systematic reviews show that XR has progressed from experimental deployments toward scalable educational applications that support authentic learning, simulation-based training and

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collaborative knowledge construction across multiple disciplines [2, 3], as they are also designed to support usability, and real-world impact [6].

This research introduces DfXR, not as a new design methodology, but as analytical lens inspired by established Design-for-X (DfX) principles widely used in engineering, where products and systems are designed based on multiple lifecycle requirements from the embryotic stages of development. Similar to Design for Manufacturing, Design for Assembly or Design for Maintenance, a DfXR perspective emphasises that immersive educational environments should be designed not solely for technological functionality but for their intended educational use, learner interaction and organisational implementation [9]. Accordingly, XR is viewed not as an isolated technological artefact but as part of an educational service system. Based on previous research the effectiveness of such a system depends upon the deliberate alignment of pedagogical design, interaction design, accessibility, technical feasibility and organisational context [10–12]. This is further acknowledged within educational technology research involving immersive learning, where immersive learning and efficient learning experiences are increasingly manifested as a design challenge [13, 14]. While research [2] identify instructional design as one of the principal determinants of educational effectiveness in immersive environments [2], another path argue that XR training should be grounded in evidence-based learning theory rather than technological novelty [15]. Similarly, other studies demonstrate that learner presence and engagement are strongly influenced by the design of immersive learning activities [16], and that successful educational implementation depends upon integrating pedagogical, technological and organisational considerations throughout the development process [17].

Despite these developments, previous research has predominantly examined learning outcomes, technological performance and user acceptance [1, 2, 6]. Comparatively limited attention has been devoted to identifying the recurring design principles that enable immersive educational services to be successfully developed, implemented and sustained across authentic organisational contexts [17, 18]. While existing research demonstrates the educational potential of XR, comparatively little guidance exists regarding how immersive learning environments should be systematically designed to support learner engagement, organisational adoption and long-term educational service innovation [2, 16, 19]. Addressing this gap therefore requires moving beyond evaluating XR technologies towards understanding how educational value is created through DfXR, where pedagogical objectives, immersive interaction and organisational implementation are treated as complementary design concerns rather than independent implementation stages. While educational services are undergoing rapid digital transformation and many times functions as learning testbeds, institutions increasingly integrate digital technologies and more precisely XR to enable flexible, learner-centred and experience-oriented models of teaching and professional development [6, 10]. In the process of this rapid transformation, immersive technologies are increasingly viewed not simply as instructional tools but as mechanisms for redesigning educational service delivery through authentic, collaborative and experiential learning environments [2, 20]. Consequently, this paper builds on the notion that authentic use-case design provides an important level of analysis because it captures how immersive technologies are translated into meaningful educational services that support engagement, skill development, and long-term adoption across diverse learning contexts [2, 21–23].

To understand how educational value emerges through intentional design decisions across the lifecycle of immersive learning environments, this study examines how authentic XR use-case design supports educational service innovation by identifying the design principles that facilitate learner engagement, user adoption, and organisational implementation. To address this aim, the following research question is investigated: What are the key design dynamics and implications of authentic XR use-cases for skills development, and how do they influence learning and professional training practices? In addition, this study contributes to the growing body of XR research in a series of distinctive ways. First, it identifies recurring design principles across authentic XR implementations that explain how immersive learning environments generate educational value. Second, it develops a comparative understanding of user adoption by integrating the perspectives of both XR developers and learners, thereby extending current knowledge beyond technology acceptance toward educational service implementation. Third, it synthesises the empirical findings into an empirically grounded DfXR perspective that informs the proposed Educational Service Design Framework and provides practical guidance for educators, instructional designers, XR developers and organisations designing and implementing immersive educational services and their potential gains. The research also provides indications on how to get started with efficient evaluation of XR efforts being made from an organisational level, although this is more a desireful next step rather than intended distinct output.

Literature Review

Educational Service Design and Immersive Learning

Rather than viewing XR solely as an emerging technology, recent educational research increasingly conceptualises immersive environments as pedagogically designed service systems in which educational value is co-created through the interaction between technological capabilities, instructional design and learner participation [2, 24, 25]. This perspective shifts attention away from technological sophistication towards understanding how immersive experiences are intentionally designed to facilitate engagement, reflection and authentic skills development. Consequently, successful implementation depends not only upon immersive hardware but equally upon educational design that aligns technological affordances with overall readiness and intended learning outcomes [16, 26, 27]. This development reflects a broader transition within educational technology research from technology-centred explanations towards learner-centred perspectives.

Earlier studies frequently evaluated XR according to technical performance and levels of immersion, whereas more recent work demonstrates that meaningful learning emerges through carefully structured interaction, authentic activities and pedagogical integration [2, 3, 16]. Factors such as presence, agency, motivation and cognitive load mediate the relationship between immersive technology and learning outcomes, emphasising the importance of instructional design in achieving educational effectiveness [19]. Systematic reviews similarly demonstrate that immersive learning environments consistently improve learner engagement, collaboration and experiential learning when embedded within authentic educational contexts rather than functioning as isolated technological demonstrations [17, 18]. These findings support viewing XR as an educational service platform capable of supporting continuous knowledge development, collaborative learning and organisational capability building. Nevertheless, considerable variation remains regarding how immersive educational experiences should be designed to support sustainable implementation across different learning contexts. This motivates a closer examination of educational service design from a use-case perspective.

Authentic XR Use-Case Design

Authentic use-cases constitute the mechanism through which immersive technologies are translated into meaningful educational experiences. Rather than emphasising technological realism alone, authentic scenarios seek to recreate professional contexts in which learners actively engage with realistic problems, collaborative decision-making and iterative experimentation [13, 21]. Such environments support experiential learning by enabling participants to apply theoretical knowledge within contexts that closely resemble future professional practice, thereby strengthening knowledge transfer and long-term competence development [15, 28].

The growing adoption of digital twins further illustrates how immersive technologies increasingly bridge physical and virtual learning environments. By integrating real-world equipment with immersive simulations, digital twins provide opportunities for learners to validate decisions, evaluate alternative solutions and observe authentic system behaviour without disrupting operational environments [29]. Particularly within engineering education, XR environments support visualisation, interactive reasoning and problem-solving that would otherwise remain difficult to reproduce using conventional educational approaches [22]. These developments extend beyond individual learning activities by supporting organisational knowledge creation and collaborative innovation across geographically distributed participants [30].

Developing effective use-cases therefore requires balancing technological capability with educational relevance. Previous studies emphasise that immersive learning environments should not simply maximise realism but instead create meaningful opportunities for learner interaction, reflection and feedback [2]. Consequently, XR design represents a multidisciplinary activity integrating spatial organisation, interaction design, visual communication, audio feedback and user experience into coherent educational scenarios [4, 12–14, 31]. Rather than functioning independently, previous research has begun to investigate how these design dimensions collectively shape a new foundation [32–34], which pay attention to learners' perceptions of presence, usability and authenticity, thereby influencing educational outcomes and organisational adoption.

Design Dimensions Supporting Educational Implementation

Although XR technologies continue to mature rapidly, widespread implementation within educational and organisational settings remains influenced by technical, pedagogical and organisational considerations. Previous research consistently identifies investment uncertainty, implementation costs, hardware availability, cybersickness and organisational readiness as factors influencing successful adoption [23, 31, 32]. However, recent systematic reviews continue to argue that these barriers should not be viewed independently but rather as interconnected design challenges requiring educational, technological and organisational alignment

[2, 17, 35]. A noticeable aspect to recent XR literature is that several recurring educational design characteristics emerge irrespective of disciplinary context. In order to get started with XR, *progressive onboarding* supports learner confidence by gradually introducing increasingly complex immersive interactions, thereby reducing technological anxiety and cognitive overload [36]. Once a learner is introduced to the features of XR, focus is to enable a stepwise progression. Here, *interaction simplicity* becomes a way to promote intuitive communication with digital content using natural interfaces, enabling learners to focus upon educational tasks rather than technological operation [12, 13]. Across age groups, and professional levels, *authentic simulation* enhances knowledge transfer by embedding learning activities within realistic professional situations that encourage experimentation and collaborative problem-solving [6, 28]. Finally, as users are learning from various XR environments, the importance of scaling and modifying is growing, as skills are developing. Thus, *design flexibility* enables immersive learning environments to accommodate diverse learner characteristics, educational objectives and organisational requirements, thereby supporting long-term implementation and sustainable educational services [17, 18, 34].

Collectively, previous research demonstrates increasing convergence around the importance of educational service design as the primary mechanism through which XR generates value. While terminology differs across studies, the literature consistently emphasises that successful implementation depends upon balancing learner engagement, authentic practice, usability and organisational feasibility. Drawing upon this synthesis, the present study adopts four recurring design dimensions, progressive onboarding, interaction simplicity, authentic simulation and design flexibility, as an analytical framework for examining authentic XR use-cases from both developer and user perspectives. By analysing how these dimensions are manifested across different educational scenarios, the study seeks to explain how immersive technologies can support educational service innovation and sustainable organisational implementation.

Methodology

This study adopts an exploratory qualitative multi-case research design to investigate how DfXR educational services are designed and implemented across industrial and higher education contexts. The study examines multiple authentic XR use-cases to identify design considerations, implementation challenges, and educational value across different application settings. Data were collected from complementary stakeholder perspectives, including designers, technology providers, and end-users, enabling analysis of the relationship between design decisions, technological implementation, and user experiences. Each use-case was documented using a common analytical structure covering educational objectives, implementation processes, technological configurations, stakeholder involvement, and post-deployment reflections. This structure enabled systematic cross-case comparison while preserving the contextual richness required for qualitative design research. Authentic operational scenarios were intentionally selected as information-rich cases, as realistic XR environments provide opportunities to examine how immersive technologies, design practices, and learning objectives are aligned through situated experiences and iterative development processes [15, 20, 28, 37]. Likewise, following the co-design perspective presented by Bailenson [38], the development process recognised that successful XR educational services require continuous negotiation between pedagogical objectives, technological possibilities, and stakeholder expectations. Consequently, the selected cases provide empirical foundations for identifying and synthesising DfXR principles relevant to educational service innovation.

Case Selection and Sampling Strategy

The study employed a purposive multi-case sampling strategy informed by the design research sampling framework [39]. Rather than seeking statistical representativeness, the objective was to identify information-rich cases capable of generating transferable design knowledge regarding DfXR educational services. Following the guiding notions by [39], case selection was guided by expected knowledge contribution, contextual variation, stakeholder diversity and analytical potential.

Initially, ten XR use-cases developed through industrial collaborations and educational initiatives were reviewed. The preliminary review served as an analytical screening stage, where each case was assessed using four predefined criteria derived from the sampling framework. Each criterion was rated on a three-point scale (1 = low, 2 = moderate, 3 = high), enabling transparent comparison while supporting qualitative judgement (Table 1).

Cases with the highest overall analytical value and complementary characteristics were selected for detailed investigation (see Table 2). Following the assessment, four use-cases were purposefully selected: two employing AR and two based on fully immersive VR. The final sample provides variation across the two dominant XR modalities while maintaining a common educational purpose, enabling analytical rather than statistical generalisation. Adopting this sampling strategy supported the subsequent cross-case comparison to further explore how different XR scenarios

Table 1 Selection criteria applied during case screening

Criterion	Assessment focus	Rating (1–3)
Authentic relevance	Extent to which the case represented realistic engineering or industrial learning activities	1 = Low, 2 = Moderate, 3 = High
DfXR design contribution	Ability to demonstrate distinct educational, interaction or implementation challenges	1 = Limited, 2 = Moderate, 3 = Strong
Technical maturity	Feasibility, stability and completeness of the implemented XR solution	1 = Prototype, 2 = Partially validated, 3 = Operational
Comparative value	Potential to contribute complementary insights across cases and XR modalities	1 = Low, 2 = Moderate, 3 = High

Table 2 Selection matrix

Case	Authentic relevance	DfXR contribution	Technical maturity	Comparative value	Selected
UC1	3	3	3	3	✓
UC2	3	2	3	3	✓
UC3	2	3	3	2	✓
UC4	3	3	2	3	✓
UC5	2	2	2	1	–
UC6	1	2	3	1	–
UC7	2	1	2	2	–
UC8	1	1	2	2	–
UC9	2	2	1	1	–
UC10	1	1	2	1	–

influence educational service design, implementation and user experience.

Participants and Data Sources

To capture the complexity of DfXR educational service development, data were collected from multiple stakeholder groups representing the design, implementation, and use of the XR applications. Consistent with recommendations for triangulation in qualitative design research [20], the study integrates evidence from designers, technology providers, educators, and end-users to obtain complementary perspectives on the development process. The primary design perspective was provided by the lead XR designer in support of technology developers by sharing both distinct use-case design procedure, but also key considerations throughout the design process. While the XR designer was responsible for the conception, development, and implementation of the four use-cases. This perspective was complemented by two technology providers who contributed technical reflections on software implementation, design decisions, deployment challenges, and practical trade-offs encountered during development. User data were obtained from engineering design students participating in XR workshops within two advanced product development courses, together with participants from industrial learning contexts. Participants were eligible if they had completed at least one of the developed XR learning activities, ensuring that all reflections were grounded in direct experience of the applications. Across the study, a dozen semi-structured interviews were conducted with end-users, the lead XR designer, and the two technology providers. Interviewees represented both higher education and industrial settings, providing variation in educational background, prior XR experience, and application context. Structured workshop reflections and observational notes collected during the design and deployment of the four use-cases provided complementary qualitative data, enabling triangulation between design intentions, practical implementation decisions and user experiences.

Data Collection and Analysis

Data were collected throughout the iterative design, implementation and deployment of the four DfXR use-cases using complementary qualitative methods. Semi-structured interviews served as the primary data source, exploring participants' experiences of interaction design, usability, immersion, implementation challenges, learning support and opportunities for improving DfXR educational services. These were complemented by structured workshop reflections completed immediately after the XR activities and observational notes documenting implementation decisions, technical constraints, user behaviours and recurring design challenges. The empirical material was analysed using an abductive qualitative approach, iteratively relating emerging patterns to existing literature on immersive learning, educational psychology, human–computer interaction and XR implementation. Interview transcripts, workshop reflections and observational notes were analysed through open, axial and selective coding. Open coding identified recurring concepts related to interaction, usability, accessibility and learning experiences; axial coding grouped

these into broader categories of pedagogical design, technological implementation, user interaction and organisational constraints, and selective coding synthesised higher-order themes through cross-case comparison. Analysis continued until thematic saturation was reached, with no substantially new themes emerging. Findings were interpreted through a comparative analysis of design intentions, implementation decisions, and end-user experiences, highlighting areas of alignment and divergence across the dataset.

Design Considerations and Contextual Constraints

The development of DfXR educational services involved the specification of use-cases addressing educational objectives, industrial contexts, accessibility requirements, and deployment considerations. The selected use-cases were structured to represent engineering workflows while maintaining suitability for educational application. This balance reflects recent literature emphasising that successful XR learning environments need a strong focus on pedagogical usability, stakeholder acceptance, and organisational feasibility [15, 20, 22, 28, 32]. Because several use-cases originated from industrial collaborations involving proprietary products and manufacturing processes, visual representations and procedural descriptions were deliberately synthesised before publication. Rather than reproducing confidential industrial scenarios, the presented cases preserve the educational and design characteristics while removing commercially sensitive information. This approach aligns with argument that design representations should support *"building shared understanding"* between stakeholders while protecting contextual confidentiality [37, p. 17]. Accessibility also constituted a fundamental design principle throughout the research. While high-fidelity XR systems offer considerable educational opportunities, they frequently require specialised hardware, advanced programming expertise, and substantial financial investment. Such requirements may limit wider adoption across educational settings. Consequently, based on use-case design narratives, the development philosophy prioritised educational accessibility by favouring low-code and commercially available development environments whenever possible without compromising educational authenticity. This design decision is consistent with current DfXR literature advocating scalable and sustainable educational services that can be implemented beyond highly specialised research laboratories. One of the VR use-cases illustrates this philosophy by demonstrating how immersive assembly training can be developed using accessible interaction paradigms while maintaining realistic learning objectives. Likewise, the AR use-cases were implemented using Microsoft HoloLens 2 together with Microsoft 365 Guides.

Reviewing the Design of XR Use-Cases

Based on XR use-cases that had been pre-developed through an iterative DfXR process in which learning objectives, system capabilities, and interaction requirements were progressively translated into functional immersive learning environments. In order to choose and target authentic use-cases, it is fundamental to also have proper hardware available, together with software capability, and deployment context. This review notion is consistent with prior XR learning research emphasising the importance of aligning technological and pedagogical design decisions [4, 6, 32, 40]. For the AR use-cases, system design combined a desktop-based authoring environment with Microsoft HoloLens 2. This configuration enabled the creation of structured instructional sequences, integration of three-dimensional engineering models, and spatial placement of digital guidance within physical environments. The resulting system supported stepwise procedural instruction in real-world workspaces, where digital content was aligned with physical objects through spatial calibration to ensure consistent registration between virtual and physical elements. Interaction was enabled through gaze, gesture input, voice commands, and QR-code-based activation for initiating predefined learning scenarios. The VR use-case was developed as a fully immersive simulation using Unity and its XR Interaction Toolkit. The system architecture followed a scene-based structure in which spatial environments, interaction logic, and object behaviours were implemented within a unified development framework. Unity's XR toolkit provided support for spatial interaction, object manipulation, and navigation within immersive environments, while C# scripting was used to define interaction logic and system behaviour.

Across both AR and VR implementations, development proceeded through iterative cycles of prototyping and refinement. Initial prototypes were developed to establish instructional flow and core interaction functionality, followed by incremental refinement of spatial alignment, interaction behaviour, and task sequencing. For AR, this included content creation in Microsoft 365 Guides, integration of instructional steps with 3D models, and calibration of holographic overlays within physical environments prior to deployment on HoloLens devices. For VR, development involved scene construction in Unity, configuration of XR interaction components, and iterative implementation and testing of interaction logic using C#. The development workflow involved translating learning objectives into structured instructional sequences, representing these sequences within spatial environments, and implementing interaction mechanisms appropriate to the selected XR modality. The process followed an iterative DfXR approach, with educational requirements,

technical constraints, and interaction design considerations being addressed throughout development. This approach is consistent with prior work that frames XR development as an integrated design and engineering activity process [20, 38].

Empirical Findings

The empirical findings are presented through four authentic XR use-cases representing different stages of immersive educational service development. Although each use-case addresses a distinct industrial application, together they illustrate how AR and VR technologies support assembly guidance, training, prototyping and collaborative design. Rather than interpreting the findings through predetermined theoretical constructs, this section presents the empirical observations emerging from each implementation.

Use-Case I: AR-Supported Mobile Phone Assembly

Scenario

The first use-case was developed within an organisational initiative to establish an internal XR test environment comprising multiple immersive training stations supporting different in-house service functions. The broader objective was to explore how XR technologies could support assembly training, design for assembly, and internal product prototyping. As an initial demonstrator, an AR assembly application was developed using Microsoft 365 Guides to investigate the feasibility of HoloLens-assisted mobile phone assembly.

Implementation Activity

Participants completed a stepwise assembly exercise in which individual mobile phone components were assembled using HoloLens. Throughout the activity, holographic overlays were superimposed onto the physical workspace, providing context-sensitive guidance for each assembly stage. Visual indicators highlighted component locations, while interactive holograms directed users through sequential assembly operations. Additional system functions included quality-control prompts, confirmation dialogues, and automated feedback indicating successful completion or identifying incorrect assembly actions. Figure 1 presents how context-sensitive holographic overlays provide visual guidance, identify component locations, support interaction with virtual assembly instructions, present quality-control prompts, and deliver automated feedback throughout the assembly process.

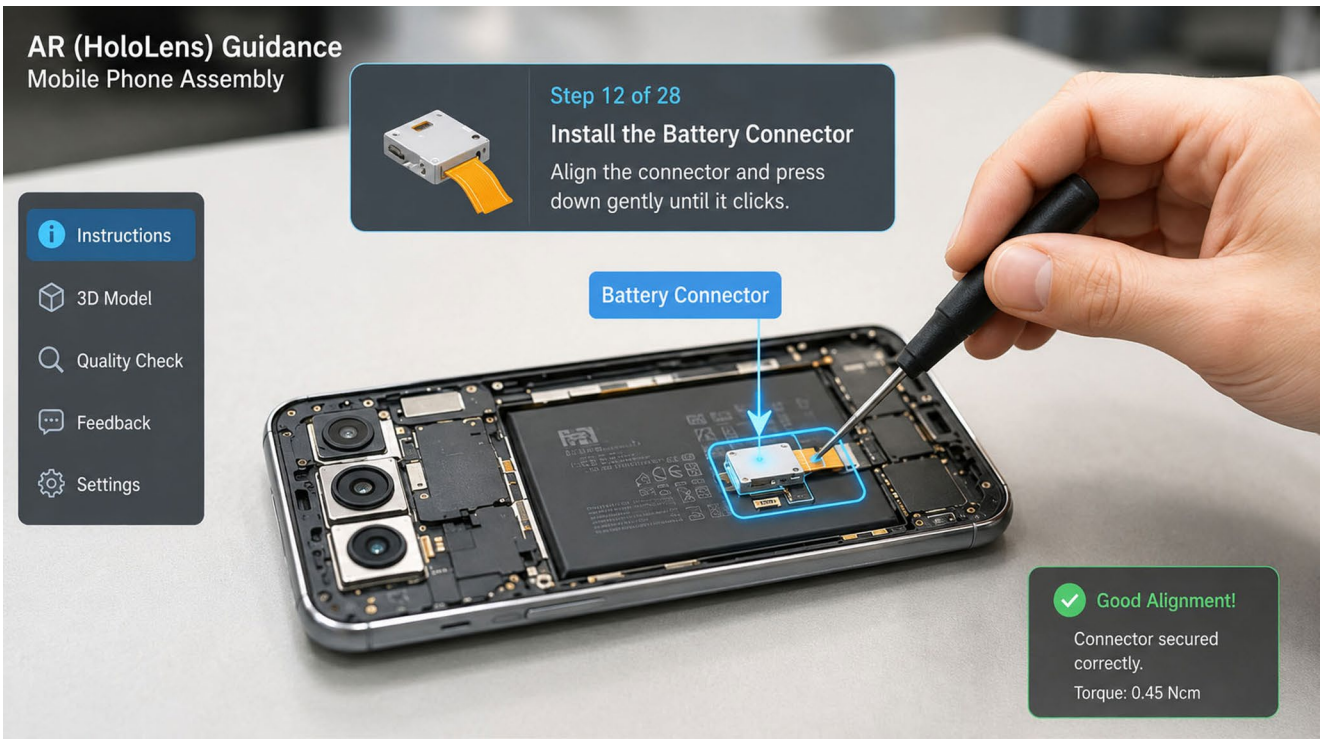


Fig. 1 HoloLens-assisted mobile phone assembly illustrating a stepwise AR navigation sequence. The figure has been modified and synthesized using chatGPT 5.5 for confidentiality reasons

Key observations

Despite having little or no previous experience of HoloLens, most participants chose not to follow the introductory tutorials describing menu navigation or gesture interaction. Instead, users relied primarily on intuitive exploration and observation of other participants before engaging with the assembly task. Although this initially created uncertainty regarding gesture recognition and interface navigation, confidence increased rapidly once participants began interacting with the holographic instructions.

The opportunity to experience the technology within an authentic assembly context encouraged participants to reflect upon its practical relevance beyond the demonstrator itself. Several users identified potential applications within their own organisations. For example, one fitter suggested adapting the AR workflow to support troubleshooting procedures during machine maintenance, illustrating how hands-on experience stimulated broader reflections regarding future workplace implementation. The lead developer similarly emphasised that allowing users to progressively explore increasingly complex assembly tasks supported both confidence and learning: *"Once participants put on the headset, they gain a practical first-hand perspective on how AR can support their existing work. Learning improves because the difficulty can be gradually increased, allowing progressively more advanced tasks to be introduced."* Overall, the observations indicate that authentic interaction with the AR environment encouraged rapid familiarisation while simultaneously enabling participants to identify practical industrial applications.

Use-Case II: AR Implementation for Industrial Assembly Training

Scenario

The second use-case focused on the organisational implementation of HoloLens within an industrial manufacturing company seeking to develop an internally customised AR solution for assembly training. The project aimed to familiarise prospective users with immersive technologies while simultaneously developing procedures capable of supporting both internal employees and external customers.

Implementation Activity

Participants were introduced to the HoloLens environment through an assembly training application designed to support new fitters during product assembly. Users navigated the system using either the standard HoloLens interface or QR-code-based launching procedures developed specifically for the project. The implementation additionally explored alternative methods for developing and distributing AR experiences across different organisational contexts.

Key Observations

Participant experiences varied according to prior familiarity with immersive technologies. Although conventional HoloLens menu navigation was available, approximately half of the participants preferred launching the application through pre-generated QR codes, avoiding the standard navigation process. Participants with limited prior experience reported greater difficulty with menu interaction, particularly when interpreting how hand gestures activated different system functions. Despite access to instructions, some participants omitted configuration steps or misunderstood gesture-based interactions. These observations indicate that initial engagement with immersive technologies involved a learning threshold, as users needed to adapt simultaneously to unfamiliar interaction conventions and complete the intended assembly tasks. The lead designer reported that the graphical authoring environment reduced perceived development complexity compared with fully programmed XR applications. This was described as supporting the creation of instructional experiences by educators with limited programming expertise. Developer observations further highlighted the importance of reducing unnecessary interaction demands during initial implementation. Guided setup procedures and simplified access mechanisms were identified as contributing factors that allowed users to focus on the learning activity rather than system operation.

Use-Case III: VR-Supported Assembly Validation

Scenario

The third use-case explored how VR could support assembly validation by assisting installers during bolt fastening operations. The objective was to investigate whether immersive simulations could improve assembly quality by helping users identify incorrectly tightened bolts while simultaneously providing a safte and realistic training environment.

Implementation Activity

Participants completed an assembly exercise involving thirty virtual bolts within a VR environment. A physical torque screwdriver was digitally tracked, allowing its position, rotation and trigger actions to be synchronised with a virtual screwdriver developed in Unity. Correctly tightened bolts generated immediate visual confirmation through illuminated green indicators together with corresponding audio

feedback. Bolts that had not achieved the required torque remained visually distinguishable, enabling users to identify incomplete assembly operations throughout the exercise. Figure 2 presents VR-supported assembly process, which illustrates the synchronised interaction between a physical torque screwdriver and its virtual representation. For full immersiveness, the combined visual and audio feedback confirm correctly completed fastening operations while supporting procedural training and quality assurance.

Key Observations

Participants generally regarded the simulation as realistic, although several reported difficulties accurately positioning the virtual screwdriver when initiating bolt fastening. In practical assembly situations, physical contact naturally guides tool alignment, whereas the current virtual implementation required users to establish this alignment manually. Both users and developers suggested that introducing stronger coupling between the physical and virtual interactions could substantially enhance the perceived authenticity of the experience. Although the simulation successfully reproduced assembly procedures, participants consistently associated increased physical realism with improved learning and greater confidence in the training environment.

Use-Case IV: Flexible VR Design Platform

Scenario

The final use-case extended the previous assembly scenario by developing a flexible VR platform supporting rapid creation and modification of immersive assembly experiences. Rather than focusing solely on end-user interaction, the project investigated how immersive environments could be efficiently developed and customised using accessible development tools requiring limited programming expertise.

Implementation Activity

The platform enabled developers to import CAD models, define assembly and disassembly sequences, and configure complete assembly workflows within a VR environment. Unlike Microsoft 365 Guides, where interaction follows predefined templates, the Unity-based platform provided considerably greater flexibility, allowing developers to customise interaction logic, assembly sequences and interface behaviour. Participants manipulated a physical screwdriver synchronised with its virtual counterpart, enabling realistic interaction throughout the assembly exercise.

Key Observations

Participants consistently highlighted the value of combining physical interaction with virtual assembly activities. Although the system did not incorporate dedicated haptic feedback, users considered the synchronised physical screwdriver sufficient to provide a convincing sense of interaction during assembly.

Developer observations nevertheless identified several opportunities for further refinement. Interface precision, interaction consistency and improved handling of assembly errors were repeatedly identified as priorities for future development. Additional interest centred upon incorporating low-code and block-programming functionality to simplify the creation of customised XR applications, thereby

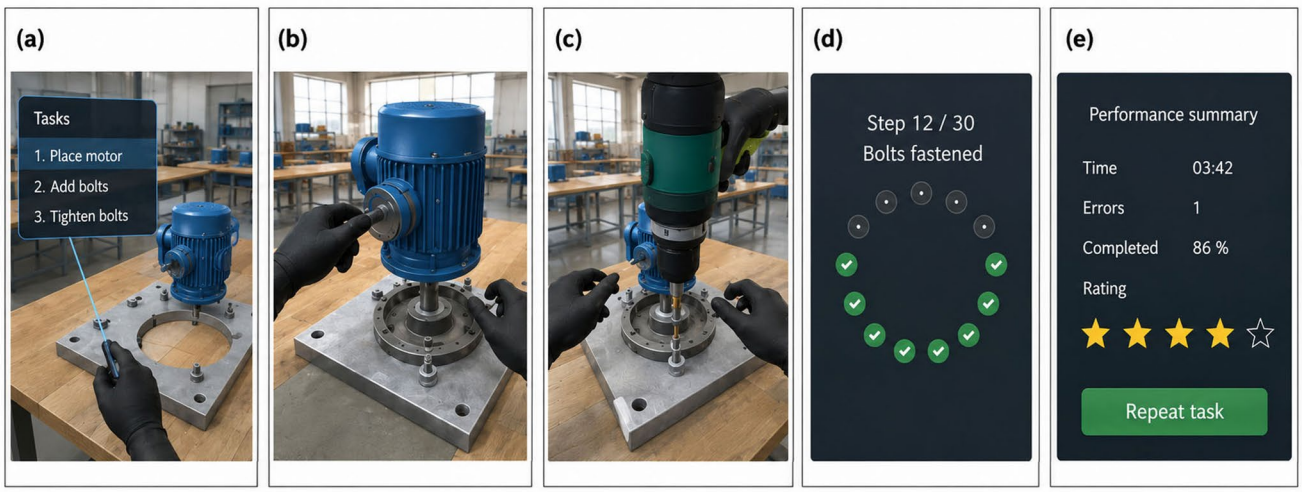


Fig. 2 Representative stages of the VR assembly task. **a** Task briefing displaying assembly objectives; **b** positioning of the motor component; **c** fastening using the virtual screwdriver; **d** real-time progress indicator showing completed bolts; **e** performance summary including completion time, errors and overall score. Note: The figure has been modified and synthesized using chatGPT 5.5 for confidentiality reasons

enabling organisations to develop and adapt immersive training scenarios without requiring advanced programming expertise. Collectively, these observations suggest that future XR platforms should balance technical flexibility with ease of authoring, allowing immersive educational services to evolve alongside changing organisational requirements while remaining accessible to non-specialist developers.

Taken together, the four use-cases reveal recurring patterns despite differences in XR technology, educational objectives and organisational context. These recurring observations suggest that successful XR implementation depends not only on technological capability but also on how immersive learning experiences are designed, experienced and integrated into organisational practice. The following analysis interprets these cross-case patterns to develop a deeper understanding of the educational and organisational mechanisms underpinning effective XR implementation.

Cross-Case Analysis

Rather than examining each principle separately, the cross-case analysis identifies the underlying educational mechanisms that emerged consistently across the four use cases. Despite differences in industrial contexts and immersive technologies, differences were found in how users interacted with virtual content, perceived simulation authenticity, learning efficiency, and adapted experiences to organisational needs. Drawing on the theoretical perspectives established in earlier Sections [15, 20, 22, 28, 32]., the analysis identifies four educational design dimensions that serve as an interpretive lens for understanding immersive learning environments rather than as predefined criteria for evaluating the cases.

Depending on user age and previous technological experience, notable differences emerged in how participants adapted to the XR system. Individuals aged 40 years and above, particularly those with limited prior exposure to immersive technologies, required more time to understand the interaction logic and frequently questioned each procedural step, whereas younger participants were generally more receptive and adapted more rapidly. These findings suggest that the primary challenge extends beyond chronological age to include cognitive barriers associated with onboarding, where novice users must simultaneously learn the assembly procedure and the interaction paradigm of the XR system. This increased cognitive demand may initially reduce confidence and efficiency until users develop familiarity with the interface and workflow [12, 15, 36]. Consequently, onboarding should not only introduce system functionality but also progressively scaffold interaction techniques to reduce cognitive load and technology resistance, particularly among first-time or less experienced users [6, 40].

Across the use-cases, XR service development demonstrates a strong connection to what earlier findings describe as the relationship between digital capability, digital innovation, and organisational performance [1]. Rather than relying solely on the availability of XR technologies, successful implementation depended on organisations possessing the competencies required to translate use-case requirements into functional XR applications, reflecting the multidisciplinary nature of XR development and the organisational readiness required for successful implementation [6, 22, 26]. A consistent pattern across the cases was that the choice of development approach should reflect both the intended application and the organisation's available competencies. While use-cases III and IV were developed in Unity to enable highly customised interactions, this flexibility required expertise in XR development frameworks, object interaction, and programming, primarily using C#, with Python providing complementary scripting capabilities through appropriate libraries and plugins. These findings reinforce that XR development extends beyond conventional software engineering by requiring specialised design and implementation competencies [12, 24, 32].

Conversely, the cross-case analysis show that limited development resources were more inclined to favour prebuilt XR solutions over bespoke development. This illustrates a recurring trade-off between technological flexibility and organisational capability, where intent and technical capacity become difficult to master. Instead, adoption is influenced not only by technical potential but also by available expertise, resources, and organisational readiness [26, 33, 40]. The use-case findings show that users with little or no prior experience with HoloLens found navigating the menus challenging and time-consuming. It emphasizes the need for intuitive design and comprehensive onboarding to facilitate user engagement. Similarly, in developing XR experiences, the balance between flexibility and ease of use is crucial. Providing clear instructions and support mechanisms can significantly enhance the user experience, whether navigating using HoloLens or creating an immersive visualization using virtual objects this need to adapt to local spatial conditions. The iterative development process, incorporating videos, images, text, audio, holograms, and interaction feedback, contributes to lowering these cognitive barriers by providing multiple complementary information channels. Features such as assembly logs further support reflective learning by enabling users to revisit completed procedures, reinforcing procedural knowledge over time. Likewise, intuitive interaction techniques, including drag-and-drop manipulation and immediate visual feedback,

reduce the mental effort required to interact with virtual objects, allowing novice users to focus on the assembly task rather than the interface itself [6, 12, 15]. While experienced users benefit from efficient interaction and review capabilities, these design characteristics are particularly valuable during onboarding, where they facilitate the transition from unfamiliar technology to confident operational use.

Discussion

The four use-cases collectively demonstrate that educational value is not created through XR technology alone but emerges through the interaction between immersive technologies, educational service design, and organisational implementation. Although each case addressed different industrial contexts and technological configurations, several recurring patterns were consistently observed across developer and user perspectives. Rather than representing isolated implementation characteristics, these recurring patterns collectively describe how immersive educational services are progressively designed, experienced, and integrated into authentic organisational practice. Drawing upon the educational service perspective developed in the literature review [2, 17, 18], four interconnected educational service design principles emerged across the cases: progressive onboarding, interaction simplicity, authentic simulation, and design flexibility. Rather than functioning independently, these principles appear mutually reinforcing, collectively supporting learner engagement, organisational adoption, and sustainable implementation of XR-enabled educational services.

Design Principles

Progressive onboarding is a key element as it emerges in XR use-cases [9, 12, 21], users consistently preferred learning through direct interaction rather than formal instruction. Despite having access to written tutorials and gesture explanations, participants typically relied upon intuitive exploration or observation of colleagues before gradually developing confidence in the immersive environment. This behaviour suggests that onboarding should not be regarded as a preliminary activity preceding immersive learning but rather as an integral component of the learning experience itself. This finding extends previous work [19], who argue that learner confidence develops through agency and progressive interaction within immersive environments. Similarly, research report that successful institutional XR implementation depends upon reducing initial technological barriers through pedagogically aligned introduction strategies [17]. The present findings contribute by demonstrating that onboarding is itself a dynamic educational design mechanism that enables users to gradually construct operational understanding while simultaneously recognising potential organisational applications.

Interaction simplicity emerged as a prerequisite for educational engagement. Participants who experienced unnecessary interaction complexity devoted considerable cognitive effort towards understanding interface behaviour instead of focusing upon the educational activity. Conversely, simplified interaction pathways, such as QR-code initiation or intuitive gesture sequences, enabled users to engage more rapidly with the learning objectives. These observations complement cognitive load theory [20], and the way research has suggested that reducing operational complexity enables learners to allocate greater cognitive resources towards authentic problem-solving rather than wasting time on technological navigation [19]. Consequently, interaction simplicity should not be interpreted merely as an interface characteristic but as a pedagogical design strategy supporting immersive learning.

Authenic simulation become critical as VR assembly scenarios demonstrated that participants consistently associated learning effectiveness with authentic task execution rather than graphical realism alone. Users accepted simplified virtual representations provided that interaction reflected realistic professional workflows. Conversely, the absence of tactile coupling between physical and virtual tools reduced confidence in procedural accuracy despite otherwise realistic visual representations. This supports situated learning literature and experiential learning theory while extending previous XR research by demonstrating that authenticity emerges through meaningful interaction rather than perceptual fidelity alone [15, 16, 28].

Design flexibility is illustrated as sustainable XR implementation depends increasingly upon organisational adaptability rather than fixed technological solutions. Developers consistently emphasised authoring flexibility, reusable components, and simplified content creation as prerequisites for long-term organisational adoption. Rather than viewing immersive applications as completed software artefacts, organisations increasingly require configurable educational service platforms capable of evolving alongside changing operational requirements. This finding aligns with recent organisational adoption research [40] while extending educational service literature by positioning XR platforms as continuously evolving service ecosystems rather than static training applications.

Table 3 Educational Service Design Framework for XR

Design principle	Educational purpose	Educational function	Organisational contribution
Progressive onboarding	Develop learner confidence	Supports gradual familiarisation with immersive interaction through authentic participation	Reduces implementation barriers and strengthens initial user adoption
Interaction simplicity	Reduce unnecessary cognitive effort	Enables learners to focus on educational tasks rather than interface operation	Improves usability, acceptance and operational efficiency
Authentic simulation	Strengthen experiential learning	Connects immersive activities with authentic professional practice and procedural competence	Enhances knowledge transfer and workplace relevance
Design flexibility	Support continuous educational adaptation	Enables customisation, iterative improvement and scalable immersive learning services	Facilitates long-term organisational implementation and continuous innovation

Towards an Educational Service Design Framework for XR

The cross-case analysis demonstrates that the educational value of XR cannot be sufficiently explained by technological capability, learning outcomes or user acceptance when considered independently. Although the four use-cases differed with respect to XR modality and educational purpose, the recurring patterns consistently highlighted that successful implementation depended upon the deliberate integration of pedagogical design, relevance and perceived quality of the authentic workplace practice. These findings reinforce recent systematic reviews, which argue that the effectiveness of immersive learning environments depends less upon technological sophistication than upon how learning experiences are intentionally designed and embedded within educational practice [2, 16, 17, 18].

Viewed through the DfXR perspective adopted in this study, these recurring observations were synthesised into an Educational Service Design Framework (see Table 3). Rather than representing a sequential implementation methodology, the framework should be interpreted as an empirically grounded conceptual synthesis that explains how complementary educational service capabilities collectively support learner engagement, authentic professional practice and sustainable organisational implementation. In doing so, the framework responds to increasing calls for multidisciplinary approaches that integrate perspectives like educational psychology, human–computer interaction, and organisational implementation within coherent XR design practices [1, 2, 6, 15, 22, 28, 40].

Unlike previous studies that have primarily evaluated individual implementation factors, such as usability, learning performance or technology acceptance [2, 6, 22], the proposed framework conceptualises these design principles as interdependent educational service capabilities. Educational value therefore emerges not from optimising individual technological characteristics, but from how complementary capabilities interact across the educational service lifecycle.

From a theoretical perspective, the framework contributes by integrating previously fragmented research traditions. It combines insights from immersive learning and educational psychology [2, 15, 20, 27]; human–computer interaction and interaction design [12, 24]; organisational implementation and digital innovation [1, 40], and emerging XR educational research [17, 18, 34, 35]. Rather than treating these perspectives as independent research streams, the framework demonstrates how they collectively explain the design of sustainable immersive educational services.

Consequently, the framework represents more than a synthesis of the empirical findings. It establishes an empirically grounded conceptual foundation for DfXR, illustrating how immersive educational services can be intentionally designed to align pedagogical objectives, learner interaction, technological capability and organisational implementation. In this respect, the framework provides a transferable analytical perspective that extends beyond the four investigated use-cases while remaining sufficiently flexible for future refinement and validation across other educational and professional contexts.

Conclusions

This study explores four authentic XR use-cases to examine how immersive technologies contribute to educational service innovation within industrial learning environments. By moving beyond technology-centred evaluations, the study demonstrates that the effectiveness of XR is fundamentally shaped by the intentional design of immersive educational services that align pedagogical objectives, learner interaction and organisational capability. The study demonstrates that sustainable educational value depends upon how immersive learning experiences are intentionally designed and integrated into organisational practice. Viewed through the DfXR perspective adopted in this study, the cross-case synthesis resulted in an empirically grounded Educational Service Design Framework comprising four

complementary design principles: progressive onboarding, interaction simplicity, authentic simulation and design flexibility. Rather than representing independent implementation factors, these principles collectively explain how immersive learning environments facilitate learner engagement, support authentic professional practice, and strengthen organisational adoption. Consequently, the study shifts the focus from viewing XR as a technological artefact towards understanding it as an educational service platform through which learning experiences are designed, delivered and continuously refined. By integrating perspectives from immersive learning, human–computer interaction, educational psychology and digital innovation, the proposed framework offers a transferable conceptual lens for designing, implementing and evaluating XR-enabled educational services across industrial and educational contexts.

Theoretical Implications

This study contributes to XR research in three important respects. First, it advances current educational XR literature by adopting a DfXR perspective that conceptualises immersive environments as educational service systems rather than technological artefacts. In doing so, the study responds to recent calls for multidisciplinary XR research by integrating pedagogical, interactional and organisational considerations within a unified design perspective, thereby shifting attention from technology-centred implementation towards the intentional design of immersive educational services. This perspective complements recent calls for multidisciplinary XR research by integrating pedagogical, interactional and organisational perspectives within a unified conceptual model. Second, the study identifies four recurring educational service design capabilities that consistently emerged across authentic industrial implementations. Unlike previous research, which frequently evaluates isolated implementation factors, the proposed framework demonstrates how these complementary capabilities collectively shape learner engagement, organisational adoption and educational effectiveness. Third, the framework provides an analytical lens that presents potential extensions beyond the empirical cases investigated here. Thus, rather than prescribing specific technological solutions, the framework offers theoretically grounded design guidance applicable across different XR modalities, educational contexts and organisational settings.

Practical Implications

Viewed through the DfXR perspective, the proposed Educational Service Design Framework provides practical guidance for organisations, educators, instructional designers and XR developers seeking to design, implement and evaluate immersive learning environments. First, onboarding should be regarded as a pedagogical design activity rather than a technical introduction, allowing learners to progressively develop confidence while engaging with authentic tasks. Second, interaction design should minimise unnecessary complexity so that cognitive effort is directed towards learning activities rather than system operation. Third, immersive scenarios should prioritise authentic professional workflows over technological novelty, thereby strengthening knowledge transfer and learner engagement. Finally, organisations should develop flexible XR platforms capable of adapting to evolving educational requirements, enabling immersive services to be continuously refined rather than treated as fixed technological solutions. Taken together, these recommendations suggest that successful XR implementation requires balancing educational objectives, user experience and organisational capability throughout the design process. Such a service-oriented perspective provides a practical foundation for developing immersive learning environments that create sustainable value for both learners and organisations.

Limitations and Future Research

The investigated use-cases represent a specific organisational setting and therefore do not capture the full diversity of educational disciplines, institutional environments or XR implementation strategies. Consequently, the transferability of the framework would benefit from validation across additional industrial sectors, higher education institutions and vocational training contexts in light of UN's sustaialbe development goals. Future research is suggested to examine the proposed design principles through longitudinal studies that investigate how they influence learner engagement, skills development and organisational adoption over time. Also, how this can be critically assessed from an SDG4 perspective, striving to enhance global considerations for quality educdation. Investigating larger number of XR modality settings could provide understandings across disciplines and how learner populations potentially would influence the framework interception across multiple educational contexts. In addition, emerging developments in artificial intelligence and adaptive XR environments present opportunities to investigate how immersive educational services can become increasingly personalised. Finally, future work should explore how the Educational Service Design Framework can be operationalised as an evaluative tool for assessing the maturity and effectiveness of XR-enabled educational services. Such research would provide valuable evidence for organisations seeking to implement immersive technologies strategically while further refining

the theoretical understanding of educational service design in XR.

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Informed Consent Participants taking part in this research have done so freely and is based on their willingness to share information crucial for pursuing the purpose and objective of the paper.

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